

Intrusion detection and defence for CAN bus through frequency anomaly analysis and arbitration mechanism

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As automobile intelligence continues to develop, the electronic control units connected to the Controller Area Network (CAN) bus within vehicles face an increasing number of threats from potential attacks originating from the internet. To address this issue, an intrusion detection and defence method is proposed that is capable of detecting illegal messages through the use of frequency anomaly detection, and subsequently disrupting them through utilization of the CAN bus arbitration mechanism. This proposed method offers a simple implementation compared to traditional approaches, while being able to identify suspicious units and neutralize threats in real-time. Experimental results demonstrate the effectiveness of this approach.

Introduction: In recent years, there has been a remarkable advancement in automotive intelligence, leading to a significant increase in the integration of sensors and processing modules within vehicles [1]. These modules are typically supervised by electronic control units (ECUs) and interconnected via the Controller Area Network (CAN) within the vehicle. Although the existing CAN protocol primarily focuses on achieving stable communication, it fails to adequately address the crucial issue of safeguarding the information security of individual nodes. Meanwhile, with the development of automotive intelligence and connectivity, vehicles have evolved from isolated entities to mobile smart network terminals [2–4]. Consequently, ECUs that were once confined to a relatively closed network environment within the vehicle are now confronted with a significantly heightened risk of external network security issues [4–6].

The existing research on security protection for vehicular CAN bus mainly includes security defence schemes based on cryptographic primitives [7–9] and intrusion detection methods [6, 10–14]. The former schemes are limited by the bandwidth of CAN bus, strong real-time requirements, and node computing performance, and need to compromise on security by transmitting authentication tags through lightweight authentication frameworks or covert channels. The latter schemes are able to detect the features used in intrusion behaviour under limited resource conditions and provide the capability to authenticate the legitimacy of the information, making it more suitable for vehicular CAN bus. However, existing researches have not fully considered the variations in legitimate CAN bus messages in the dynamic vehicular environment, which can lead to missed detections and false alarms. Additionally, there is a lack of effective methods for disrupting illegal messages.

Therefore, we propose a method for intrusion detection and defence for CAN bus based on frequency anomaly analysis and arbitration mechanism. This method detects the transmission frequency of legitimate messages, determines the presence of abnormal messages, and achieves long-term stable detection through online updating of discriminant parameters. Additionally, this method utilizes the arbitration mechanism of the CAN bus to promptly disrupt the transmission of suspicious messages.

Intrusion detection: The vast majority of messages on the vehicular CAN bus are transmitted by ECUs at fixed intervals [4, 6, 11, 14]. However, it is important to note that the transmission intervals for messages with different IDs can vary, and it is possible for a single ECU to transmit multiple messages with different IDs. Due to the variation in transmission delay and the deviation in message cycle, the actual time at which the intrusion detection and defence (ID&D) device connected via the vehicle's OBD-II interface receives the k th message with ID n is denoted as $t_{nk} = kT_n + d_k$, as shown in Figure 1, where T_n represents the ideal interval between messages with ID n , and d_k represents the sum of the interval deviation and transmission delay.

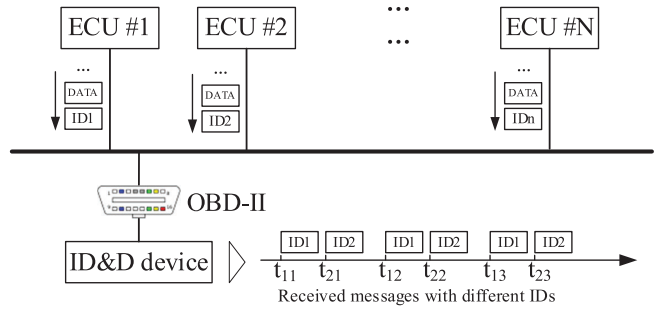


Fig. 1 ID&D device accesses the vehicular CAN bus to receive messages transmitted by each ECU

Algorithm 1 Intrusion detection

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1: Initialize:  $\mu_{1..N}, \sigma_{1..N}$  for  $N$  IDs
2:  $\mu_n \leftarrow \text{average}(T_{n1} + T_{n1} + \dots + T_{nM}), \sigma_n \leftarrow \text{std}(T_{n1} + T_{n1} + \dots + T_{nM})$ 
3:
4: function Detection ( $t_{\text{new message}}$ )
5:   if ID $_{\text{new message}}$  not in  $N$  IDs then
6:     return UnknownIDs
7:   else /* assume new message ID is  $k$ 
8:     if  $T_{\text{new message}} \in [\mu_k - \lambda\sigma_k, \mu_k + \lambda\sigma_k]$  then
9:        $T_{k \text{ random}(1, M)} \leftarrow T_{\text{new message}}$ 
10:      update  $\mu_k, \sigma_k$ 
11:     return legalMsg
12:   else
13:     return illegalMsg
14:   end if
15: end function

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By recording messages of ID n M times, we can calculate the mean μ_n and variance σ_n of the actual sending interval T_n . Upon receiving a new message with ID n , the legitimacy of the message can be determined by calculating the corresponding transmission interval T_{new} and checking if it falls within the range $[\mu_n - \lambda\sigma_n, \mu_n + \lambda\sigma_n]$. Typically, the value of λ is set to 3.

The dynamic vehicular environment can lead to gradual changes in the transmission periods of ECUs' messages. In order to ensure long-term effectiveness of detection, it is necessary to dynamically update the statistical parameters of the transmission periods for each identifier (ID) message. A typical method is to utilize the most recent M valid message reception times for each ID and calculate the mean and variance of the actual intervals [11]. In practical usage, there is a risk of rapid parameter updates leading to ineffective detection with the limited values of M . On the other hand, if the value of M is too large, it can lead to slow parameter updates, which can result in the detection algorithm not being able to adapt to changes in the transmission intervals, potentially reducing its effectiveness.

This letter adopts a stochastic policy-based parameter updating method, allowing for dynamic updates without altering the value of M . When updating a new record, instead of discarding the oldest record, we randomly replace one of the data. As a result, after N updates, the probability of a specific record still existing is $(\frac{M-1}{M})^N$. In contrast, the traditional approach results in any given record disappearing after M updates. By implementing this strategy, the dynamic updating process can effectively preserve a greater amount of historical information, thereby avoiding excessive parameter updates that could potentially lead to false detections.

The proposed intrusion detection method is summarized in Algorithm 1.

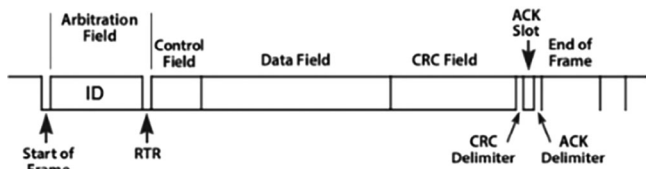


Fig. 2 A CAN 2.0A ("standard CAN") data frame where the arbitration field is located at the beginning

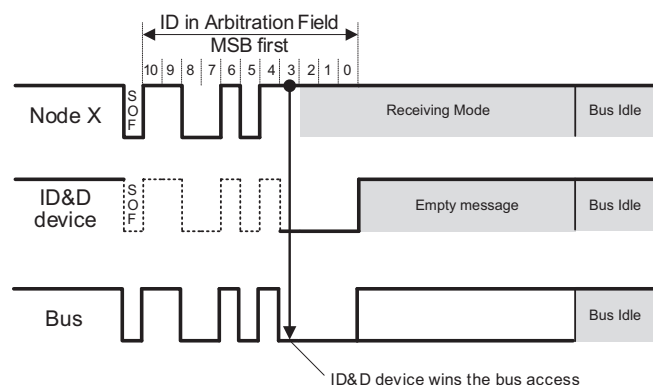


Fig. 3 The transmission of the illegal message from Node X is disrupted by a higher priority message

Disruption of illegal messages: When an unauthorized message transmission is detected on the vehicular CAN bus, we utilize the arbitration mechanism of CAN bus to interrupt the transmission, thereby reducing the impact caused by unauthorized messages.

When two or more nodes simultaneously transmit data on the CAN bus, they undergo arbitration based on the IDs in the frame's arbitration field as shown in Figure 2. The lower the ID value, the higher the priority of the corresponding message. The node with the higher priority continues transmitting, while the other nodes immediately cease transmission and enter the receive state.

Upon monitoring a message on the CAN bus, we can determine it to be an illegal message before the completion of its transmission of the arbitration field if any of the following conditions exist:

- By examining the transmitted arbitration field ID bits, the ID does not match any known valid messages' IDs.
- The intervals between this message and all valid messages are outside the normal range.
- Based on the transmitted arbitration field ID bits, the interval between this message and the potential valid messages with the same ID is outside the normal range.

After determining that the currently transmitted message is an illegal message, the ID&D device promptly transmits a message with a higher priority, that is, a lower ID value compared to the current message, to disrupt the transmission of the illegal message, as shown in Figure 3. If the detection of an illegal message occurs only after the completion of the entire arbitration field, the proposed method will be unable to disrupt the transmission of the illegal message.

Experimental result: To implement the intrusion detection and defence method proposed in this study, we employed an FPGA development board as a prototype of the ID&D device. The algorithms proposed in this study, as well as the comparative algorithms, have been implemented on the development board. The experiment also incorporated multiple ECU message simulation transmission modules and an OBD-II instrument capable of reading ECU messages from the CAN bus, forming a simulated model of the vehicular CAN bus, as shown in Figure 4.

We have configured the three ECU message simulation modules to send messages with different intervals. Specifically, module #1 is set to transmit messages with ID 38A at intervals of 10 ms, module #2 is set to transmit messages with ID 0BA at intervals of 25 ms, and module #3 is set to transmit messages with ID 2B6 at intervals of 500 ms. Then

OBD-II instrument capable of reading ECU messages from the vehicle CAN bus

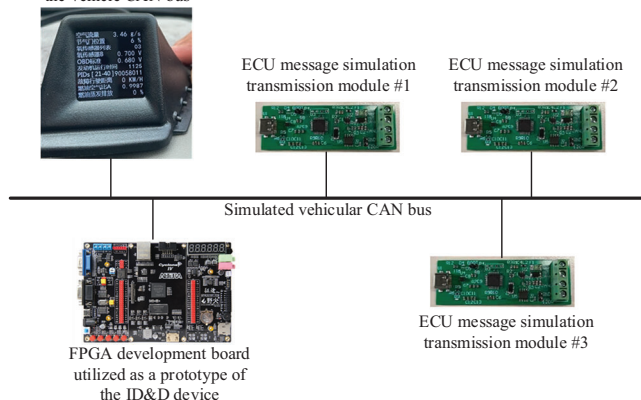


Fig. 4 The composition of the simulated vehicular CAN bus in the experiment

Table 1. Comparison of different methods in intrusion detection and defence experiments

	Our method	Our method with FIFO replacement policy ^a	Clock IDS ^b	SVM, $N = 200$ ^c	ErrIDS ^d
Recognition rate for intrusion behaviours	94.82%	92.36%	96.77%	85.45%	97.21%
Disruption rate for intrusion behaviours	79.73%	78.09%	/	/	/

^a The update strategy for intrusion detection parameters in this method employs the conventional first-in, first-out (FIFO) approach;

^b The method is described in [11];

^c The method is described in [12], where N represents the number of data frames required for fingerprint extraction;

^d The method is described in [6].

we simulated intrusion behaviours such as spoofing attacks and masquerade attacks by randomly selecting a module to transmit messages with unknown IDs, sending messages with IDs from other modules, and adjusting the transmission intervals of the original messages. Over the course of a 24-h experiment, approximately 1.2 million legitimate messages and 20,000 intrusion messages were sent. The experimental results of the proposed method and its comparison with the methods described in [11, 12] are presented in Table 1.

The experimental results demonstrate that the proposed method achieves an advanced level of recognition rate while also possessing a high probability of disrupting intrusion messages.

Conclusion: This study presents a novel intrusion detection and defence method that enables rapid detection of abnormal messages on the vehicle's CAN bus and allows for the disruption of illegal message transmission. The proposed method detects illegal messages solely based on the transmission time, thereby offering the advantage of high real-time processing capability. The detection parameter updates are implemented using a stochastic policy-based method, ensuring that a larger amount of historical information can be preserved within a fixed-size dataset. Additionally, the study also introduces an approach that utilizes the CAN bus arbitration mechanism to disrupt the transmission of illegal messages, which will effectively mitigate the adverse impact of illegal messages. The experimental results have indeed confirmed the effectiveness of the proposed method. However, to ensure its robustness and reliability, the stability of the method will be further validated in real-world application scenarios.

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review and editing. **Haoshu Shao:** Validation; visualization; writing—review and editing.

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